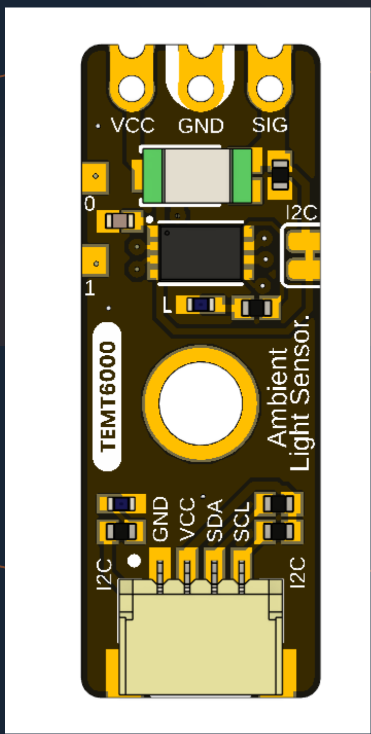
PRODUCT REFERENCE MANUAL

UNIT ELECTRONICS · DEVLAB ECOSYSTEM · ATOM FAMILY

UNIT ATOM TEMT6000

Version 1.1.0



TEMT6000 ambient-light module with an onboard I2C interface, Qwiic expansion, and direct analog access.

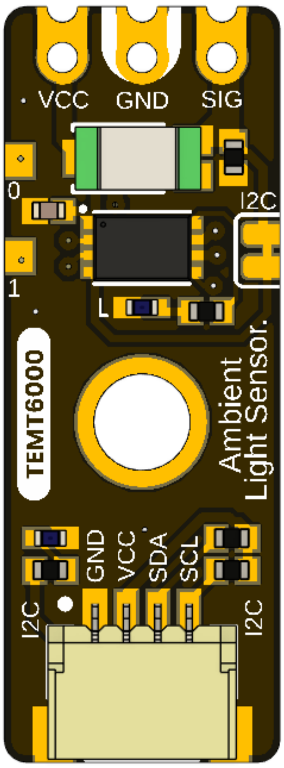
TEMT6000 | I2C + ANALOG | QWIIC | DEVLAB ECOSYSTEM | ATOM FAMILY

DOCUMENT MAP

Contents

Description

The UNIT ATOM TEMT6000 is an I2C-compatible ambient-light sensor module from UNIT Electronics, the company that creates and develops the DevLab board ecosystem. Atom is a product family within that ecosystem. Hardware V0.3.1 combines a TEMT6000 visible-light phototransistor with an onboard interface controller, Qwiic I2C routing, and direct access to the sensor's analog signal. The fitted sensor manufacturer and exact orderable suffix have not been confirmed.



The released pinout identifies two optional Qwiic connection positions, direct VCC/GND/`SIG` contacts, reserved `PA0` and PA1 pads, factory-only reset and SWD functions multiplexed on the I2C port, power and built-in indicators, and a solder bridge that disables the I2C function when cut. Inside the controller, the sensor ADC input is PA2, digital input is PA4, `BUILTIN` is driven by PB5, and the I2C bus uses `PB6/SCL/SWCLK` and `PA10/SDA/SWDIO`.

Applications

- I2C-connected ambient-light acquisition
- Direct analog light measurement during development and calibration
- Automatic display and indicator brightness control
- Day/night and relative-light detection
- Environmental data logging
- Educational I2C, ADC, and phototransistor experiments

DevLab Format Compatibility

As an Atom-family product within the DevLab ecosystem, the TEMT6000 follows the compact DevLab integration format for rapid prototyping and reuse across sensor and interface systems. The standardized layout preserves direct analog VCC/GND/SIG access while adding controller-based I2C operation.

The module provides two optional positions for a horizontal 4-pin, 1.00 mm pitch JST/Qwiic connector carrying GND, VCC, SDA, and SCL. Mechanical format compatibility does not override electrical requirements: the module supports nominal 3.3 V and 5 V operation, but the host must tolerate the actual I2C pull-up voltage or use level translation. Verify connector population and orientation, addresses, pull-ups, and bus loading.

Hardware Features

- TEMT6000 ambient-light phototransistor; fitted manufacturer/orderable part not confirmed
- 32-bit Arm Cortex-M0+ controller, up to 32 MHz
- Nominal 3.3 V and 5 V operation; controller upper operating limit 5.5 V
- 7-bit I2C slave operation from 100 kHz through 400 kHz
- Qwiic GND, VCC, SDA, and SCL routing
- Direct analog SIG access with adjacent VCC and GND
- Reserved PA0 and PA1 pads with no current application assignment
- Internal sensor ADC connection to controller pin PA2
- Internal PA4 input and shared PB6 / SCL / SWCLK / PA10 / SDA / SWDIO mapping
- Factory-only SWD/reset functions, power LED, and PB5-controlled built-in LED
- Cuttable solder bridge for disabling I2C operation
- Manufacturer Part Number (MPN) UE0098

The controller implements DevLab Device Protocol (DDP) v1.0. The TEMT6000 profile is identified by Device ID `0x0102`; command `TEMT6000_RAW` (`0x80`) returns a 12-bit ADC sample in an unsigned 16-bit little-endian response. The current controller firmware reports factory address `0x20`, firmware/hardware 1.0, and capabilities `0x000001BB`. The technical package does not provide the exact controller memory/package option, current-revision schematic, or complete module electrical limits.

SWDIO shares physical pin PA10 with SDA, and SWCLK shares PB6 with SCL; there is no separate SWD port and the two modes are mutually exclusive. SWD/reset are reserved for manufacturer programming and advanced factory diagnostics, not user firmware replacement.

1 The Board

The V0.3.1 design supports two host paths. A Qwiic-capable controller can attach through the I2C positions, while an ADC-capable host or test instrument can sample the exposed `SIG` contact directly. Factory debug reuses the same controller pins and physical port as I2C; it is not a separate user interface.

1.1 Accessories

No accessory bundle is specified. Typical integration items are:

Accessory	Purpose	Selection notes
Qwiic cable	Connects GND, VCC, SDA, and SCL	Verify 1.0 mm pitch, orientation, and contact order; connectors are shown as optional
I2C-capable host	Scans and communicates with the module	Use 7-bit addressing and a clock from 100 kHz through 400 kHz
Analog test lead or carrier	Accesses VCC, GND, and `SIG`	`SIG` must connect to a voltage-compatible ADC input
Logic analyzer	Checks I2C activity	Use input thresholds compatible with the powered board
Reference lux meter	Supports optical calibration	Required for quantitative illuminance validation

1.2 Board Identification

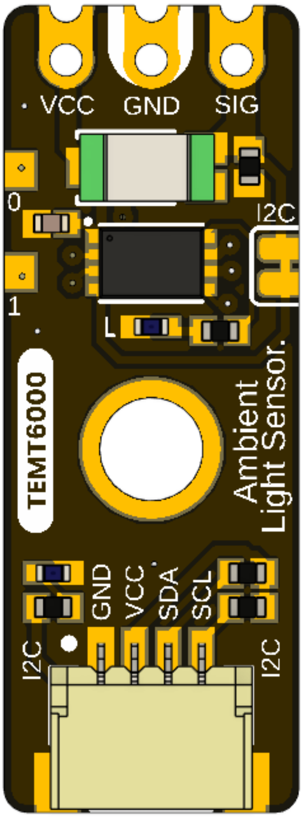
Item	Value
Product	UNIT ATOM TEMT6000 Ambient Light Sensor
Brand / company	UNIT Electronics
Board ecosystem	DevLab
Product family	Atom
Product type	I2C-compatible and direct-analog ambient-light module
Optical component	TEMT6000; fitted manufacturer/orderable part not confirmed
Interface controller	32-bit Arm Cortex-M0+ x7 variant; exact memory density and package not recorded
Manufacturer Part Number (MPN)	UE0098
Current board artwork	V0.3.1
Available schematic	Legacy analog hardware V0.0.1 only
Product Reference	Version 1.1.0

Board, pinout, schematic, and documentation revisions are controlled independently.

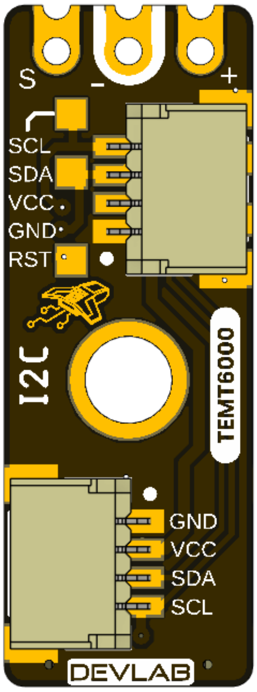
1.3 Identified Assemblies

Assembly	Function	Source status
TEMT6000 sensor	Converts visible light to photocurrent	Functional identity shown by board/pinout artwork; manufacturer and exact suffix unconfirmed
Interface controller	Samples or processes the sensor for I2C access	32-bit Arm Cortex-M0+ x7 variant; exact memory density and package not recorded
Qwiic positions A and B	I2C power and bus access	Shown as optional horizontal JST connectors
Direct contacts	VCC, GND, and analog `SIG`	Identified on the top view
I2C disable bridge	Disconnects or disables I2C when cut	Function identified; exact circuit unspecified
Power and built-in LEDs	Power and firmware indication	`BUILTIN` is driven by controller PB5
PA0/PA1 and service functions	Reserved GPIO and factory reset/debug	`PA0`/PA1 have no current application; SWD aliases share the I2C port
Internal controller mapping	PA2 ADC, PA4 input, PB5 built-in actuator, `PB6`/`SCL`/`SWCLK`, `PA10`/`SDA`/`SWDIO`	Current firmware/hardware mapping

1.4 Board Views



The top view shows the direct sensor contacts, controller, sensor, mounting hole, indicator circuitry, and one Qwiic connector position.



The bottom view labels Qwiic, the SWD aliases on that same port, reset, and the second optional connector position.

1.5 Handling

Use normal ESD precautions. Keep the transparent sensor package clean and optically unobstructed. Remove power before changing connectors or modifying the I2C solder bridge. Do not attach an SWD probe or attempt to replace the firmware. SWD/reset access is reserved for the manufacturer and advanced factory diagnostics; I2C must be inactive and isolated before factory SWD use.

2 Ratings

No current-revision schematic or released board-level electrical table is present for V0.3.1. This chapter therefore separates the expected Qwiic integration domain from comparative TEMT6000X01 reference data and explicitly marks current module values that still require release. That external data does not identify the manufacturer or exact part fitted to the board.

2.1 Current Module Operating Conditions

Parameter	Value	Status
Nominal module supply	3.3 V or 5 V	Supported operating points; use a current-limited source during bring-up
Controller operating voltage	x7: 2.0–5.5 V	Selected product variant; supports nominal 3.3 V and 5 V
I2C logic domain	Follows the actual bus pull-up rail	At 5 V, use a 5 V-tolerant host or bidirectional level translation
I2C bus speed	100 kHz to 400 kHz	Supported operating range
I2C addressing	7-bit slave	Valid configurable addresses are `0x08..0x77`
Factory I2C address	`0x20`	Pass the 7-bit value, not wire byte `0x40`
Device protocol	DDP v1.0	Command transaction followed by an exact-length read transaction
Logical Device ID	`0x0102`	TEMT6000 identity; independent of I2C address
Firmware / hardware	1.0 / 1.0	Current observable controller profile
Capability bitmap	`0x000001BB`	I2C config, digital input, analog input, sensor data, relay, watchdog, persistent config
Raw digital sample	`0` to `4095`	12-bit ADC code returned as unsigned 16-bit little endian
ADC update interval	Approximately 20 ms	Background acquisition; I2C read returns latest published sample
Command processing delay	2 to 5 ms	Use 5 ms conservatively before reading
Pending setter timeout	250 ms	Parameter must arrive before expiry
Direct `SIG` range	Not specified	Requires current analog-stage schematic and measurement
Module current consumption	Not specified	Controller, LEDs, pull-ups, and sensor load are undocumented
Module ambient temperature	Not specified	No complete-module qualification supplied

Nominal 5 V operation is supported by the controller; do not exceed its 5.5 V upper operating limit. The direct `SIG` path and I2C levels are referenced to the powered system, so every attached host must tolerate the resulting voltage or use level translation. Complete-module absolute maximums still require the V0.3.1 schematic and qualification.

2.2 Interface Controller Characteristics

The interface controller is a PY32F003 x7 variant. The following characteristics apply to this selected controller variant. The exact memory density and package fitted to this module have not yet been recorded.

Feature	Controller capability
CPU	32-bit Arm Cortex-M0+, up to 32 MHz
Memory	Up to 64 KB Flash and up to 8 KB SRAM
Operating voltage	2.0–5.5 V
ADC	One 12-bit ADC, up to 10 external channels, conversion range `0..VCC`
I2C	Standard mode 100 kHz, Fast mode 400 kHz, 7-bit addressing
GPIO	Up to 18 I/Os, all available as external interrupts
Timers	TIM1, TIM3, TIM14, TIM16, TIM17, LPTIM, IWDG, WWDG, SysTick, IRTIM
Other interfaces	SPI, two USARTs, DMA, RTC, CRC-32, two comparators, UID; SWD is reserved for factory use on this module
Operating temperature	−40 to +105 °C

The controller range establishes that 5 V is valid. It does not by itself establish pull-up resistance, complete-module current, or the absolute maximum of every external contact.

2.3 Reference-Only TEMT6000X01 Maximum Ratings

The values below come from Vishay document 81579 and are included only as a comparative TEMT6000X01 profile. They are not proof that a Vishay part is fitted, are not guaranteed for the fitted sensor, and do not define limits for the V0.3.1 controller, LEDs, pull-ups, connectors, or complete board.

Parameter	Symbol	Value	Unit
Collector-emitter voltage	`VCEO`	6	V
Emitter-collector voltage	`VECO`	1.5	V
Collector current	`IC`	20	mA
Power dissipation at 25 °C	`PV`	100	mW
Junction temperature	`Tj`	100	°C
Component operating temperature	`Tamb`	−40 to +100	°C

2.4 Reference-Only TEMT6000X01 Characteristics

Unless noted otherwise, these are values published in Vishay document 81579 at 25 °C and are provided for comparison only. Production specifications must come from the confirmed fitted part and module-level validation.

Parameter	Test condition	Min.	Typ.	Max.	Unit
Collector dark current	`VCE = 5 V`, `Ev = 0`	—	3	50	nA
Collector light current	`Ev = 20 lx`, CIE illuminant A, `VCE = 5 V`	3.5	10	16	µA
Collector light current	`Ev = 100 lx`, CIE illuminant A, `VCE = 5 V`	—	50	—	µA
Collector-emitter capacitance	`VCE = 0 V`, `f = 1 MHz`, dark	—	16	—	pF
Collector-emitter saturation voltage	`Ev = 20 lx`, `IPCE = 1.2 µA`	—	0.1	—	V
Angle of half sensitivity	—	—	±60	—	degrees
Peak sensitivity wavelength	—	—	570	—	nm
Spectral bandwidth at half sensitivity	—	440	—	800	nm

2.5 Legacy Analog Circuit Scope

The V0.0.1 schematic shows a TEMT6000 with a 10 kΩ emitter resistor, giving the first-order relation `VSIGNAL ≈ IPCE × 10 kΩ` outside saturation. Applying the 100 lx typical current from the reference-only datasheet predicts about 0.50 V; this is not a guaranteed current-board value.

V0.3.1 adds an interface controller and other circuitry. Without its schematic, the legacy resistor value and transfer function must not be represented as a guaranteed current-board characteristic. Direct `SIG` must be measured and validated on V0.3.1.

2.6 Unspecified Current-Module Characteristics

- Complete-module absolute maximum, current consumption, and power-up behavior
- Exact oscillator/timing tolerance and electrical bus-loading limits
- Pull-up resistance, bus capacitance allowance, and level compatibility
- Direct analog transfer function, load resistance, range, accuracy, and source impedance
- Guaranteed lux range, accuracy, repeatability, response time, and calibration
- Exact controller memory/package option, released firmware image, and update procedure
- Board-level temperature, ESD, EMC, humidity, and ingress ratings

2.7 Electrical Precautions

1. Use a current-limited 3.3 V or 5 V supply for engineering bring-up.
2. Verify connector orientation and share ground before applying power.
3. Do not exceed the controller upper operating limit of 5.5 V.
4. At 5 V, confirm host tolerance at `SDA`, `SCL`, and `SIG`, or add suitable level translation before connection.
5. Do not connect to reset or SWD; these signals are reserved for manufacturer programming and advanced factory diagnostics, not user firmware replacement.
6. Remove power before cutting or reworking the I2C solder bridge.
7. Confirm the current schematic and interface specification before production.

3 Functional Overview

The TEMT6000 sensor generates a light-dependent analog signal. That signal is exposed at `SIG` and is acquired by the onboard controller through ADC input PA2 for DDP access. Controller output PB5 drives the `BUILTIN` LED. The missing V0.3.1 schematic must still confirm the passive circuitry and electrical characteristics around these mapped signals.

3.1 Functional Paths

```
                                          ┌───────────────→ direct SIG contact
Ambient light → TEMT6000 ─────┤
                                          └→ PA2 ADC · controller → PA10/SDA · PB6/SCL → host
                                                         │
                            PB5 → BUILTIN · RESET

                       factory SWD aliases: PA10/SWDIO · PB6/SWCLK
                       (same physical pins as I2C; mutually exclusive)

                     PA0 / PA1 → reserved, no current application
```

3.2 Board Topology

Region	Elements	Function
Direct-contact end	`VCC`, `GND`, `SIG`, `PA0`, `PA1`	Direct analog access and reserved GPIO pads
Upper center	TEMT6000 sensor and controller	Optical sensing; internal ADC acquisition on PA2
Center	Mounting hole and product identification	Mechanical attachment and identification
Qwiic end	Indicators, bus routing, optional connector	Host power and I2C communication
Bottom side	Qwiic, reset, and SWD aliases	Alternate connector; I2C/SWD share the same physical port

3.3 I2C Mode

With the I2C bridge intact, the onboard controller is connected to the module's I2C path. Either populated Qwiic position can route `GND`, `VCC`, `SDA`, and `SCL`; the pinout labels connectors as optional. The module operates as a 7-bit slave at bus clocks from 100 kHz through 400 kHz. In normal operation, `PA10` and `PB6` are used as SDA and SCL; the factory-only SWD aliases on those pins cannot operate simultaneously with I2C.

The onboard controller firmware uses DevLab Device Protocol v1.0. A host writes one command byte, issues STOP, waits 2 to 5 ms, and then performs a separate exact-length read. Discovery reads protocol version, Device ID, firmware version, hardware version, and capabilities before any feature command is used. The TEMT6000 is identified by Device ID `0x0102`; `TEMT6000_RAW` (`0x80`) returns a 12-bit ADC code (`0` to `4095`) as unsigned 16-bit little endian. Common `READ_ADC0` (`0x60`) returns the same published PA2 sample.

The factory 7-bit address is `0x20`; configurable values are `0x08..0x77`. Current firmware/hardware versions are 1.0 and capabilities are `0x000001BB`. Hosts must still verify DDP identity rather than recognizing the product from its address alone.

3.4 Direct Analog Mode

The top-side SIG contact provides direct sensor access for ADC experiments, production test, or calibration. Its V0.3.1 transfer function and safe input range are not documented. Configure the host as a high-impedance analog input and measure the actual range before connecting a lower-voltage ADC.

The controller samples this sensor path internally on PA2, mapped to logical ADC0. READ_ADC0 and the module-specific TEMT6000_RAW command are equivalent in the current firmware.

3.5 I2C Disable Bridge

The released pinout instructs the user to cut a solder bridge to disable I2C. The missing current schematic leaves the exact isolation boundary unspecified. Do not assume that cutting the bridge removes pull-ups, disconnects every controller pin, or changes the analog transfer in a particular way. Rework only with power removed and after continuity checks on the target revision.

3.6 Optical Response

The reference-only Vishay document 81579 specifies a 570 nm peak, a 440 nm to 800 nm half-sensitivity spectral bandwidth, and a ±60° half-sensitivity angle for its TEMT6000X01. These values are comparative, not guaranteed for the unconfirmed fitted part. Source spectrum, temperature, sensor spread, enclosure windows, and mechanical shadowing can also change the response.

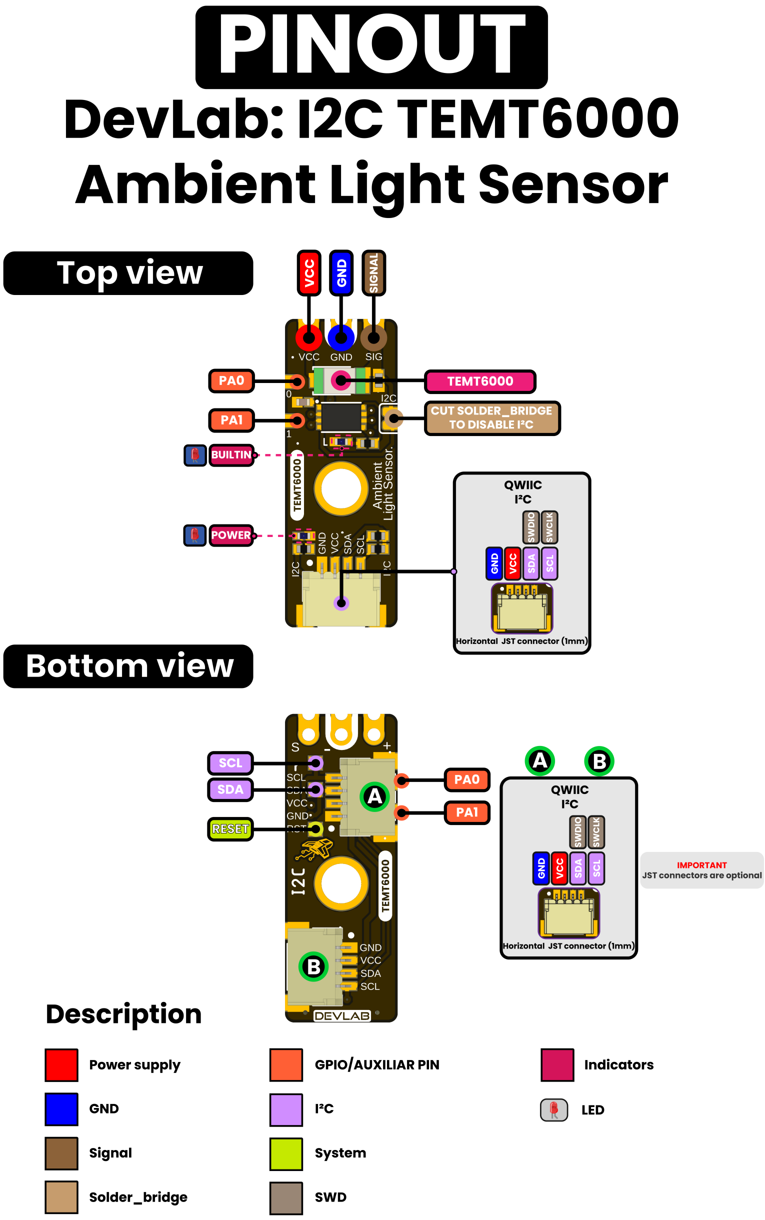
3.7 Service and Expansion Signals

The pinout exposes PA0, PA1, RESET, SWDIO, and SWCLK. PA0 and PA1 have no application assignment in the current firmware and are reserved for future use. PA4 is logical read-only GPIO0; PB5 is assigned internally to the BUILTIN LED through the relay-compatible command block. SWDIO/SWCLK are alternate functions of the same PA10/PB6 pins used for SDA/SCL, not a separate port. They and RESET are reserved for manufacturer programming and advanced factory diagnostics; user firmware replacement is unsupported.

4 Connectors & Pinouts

Pin labels below come from the released English pinout. It does not provide controlled contact numbers or a mating-connector part number, so the signal order must be checked against connector orientation before producing a harness.

4.1 General Pinout



4.2 Direct Sensor Contacts

Label	Type	Description
VCC	Power	Module supply; nominal 3.3 V or 5 V operation
GND	Power	Common power and signal reference

Label	Type	Description
SIG / SIGNAL	Analog	Direct light-dependent sensor signal; current transfer function unspecified

These three contacts are shown at the end opposite the primary Qwiic connector.

4.3 Qwiic I2C Positions A and B

Signal	Type	Description
GND	Power	Common return
VCC	Power	I2C peripheral supply; nominal 3.3 V or 5 V
SDA / SWDIO	Bidirectional I2C / factory debug data	One physical PA10 line; normal use is SDA; host must tolerate the bus pull-up voltage
SCL / SWCLK	I2C clock / factory debug clock	One physical PB6 line; normal use is SCL at 100 kHz to 400 kHz

The pinout shows two possible horizontal 1.0 mm JST connector positions and marks JST connectors as optional. Verify which position is populated on the ordered assembly. Parallel connectors are intended for bus pass-through, but that topology must be confirmed by the current schematic.

I2C and SWD are multiplexed on this same physical port: `PA10` is `SDA/SWDIO`, and PB6 is `SCL/SWCLK`. They are mutually exclusive, and there is no independent SWD connector or pin pair.

4.4 Auxiliary and Factory Service Functions

Label	Documented role	Qualification status
PA0	GPIO / reserved pin	No current application assignment; electrical limits unspecified
PA1	GPIO / reserved pin	No current application assignment; electrical limits unspecified
RESET	System reset	Factory service only; not intended for routine user operation

Do not connect an SWD probe or drive reset/debug functions. They are not user interfaces, and the product is not designed for user firmware replacement. Manufacturer diagnostics must make I2C inactive and isolate other bus devices before selecting SWD on the shared pins.

4.5 Indicators and I2C Bridge

The pinout identifies a `POWER LED`, a firmware-controlled `BUILTIN LED`, and a cuttable bridge used to disable I2C. Controller pin PB5 drives `BUILTIN`. Indicator polarity/current and bridge net connections still require the current schematic. The sensor ADC input is mapped internally to controller PA2; it is not an additional external contact in this pinout.

5 Board Operation

The repository supports safe electrical bring-up through I2C scanning, DDP identity and measurement reads, and direct analog observation.

5.1 I2C Bring-up

1. Inspect the board revision, connector population, and I2C bridge.
2. With power off, select a nominal 3.3 V or 5 V supply and connect GND, VCC, SDA, and SCL in the documented orientation. For 5 V operation, verify host tolerance at the bus pull-up voltage or add level translation.
3. Power from a current-limited source and check for unexpected heating.
4. Select an I2C clock from 100 kHz through 400 kHz; maintained examples use 400 kHz. The ESP32 and MicroPython examples map host GPIO6 to SDA and GPIO7 to SCL. Start at factory 7-bit address `0x20`, or scan if it was changed.
5. Record the active address, protocol, firmware, hardware, and capability values reported during discovery.
6. Verify protocol/firmware/hardware 1.0 and capabilities `0x000001BB`, then read either `READ_ADC0` (`0x60`) or equivalent `TEMT6000_RAW` (`0x80`).
7. Use a logic analyzer to capture start, address, acknowledge, STOP, and read timing if enumeration fails.

The maintained DDP clients are `software/examples/i2c/cpp_examples/ddp_temt6000/ddp_temt6000.ino` and `software/examples/i2c/micropython/ddp_temt6000.py`. The MicroPython example imports the reusable `software/examples/i2c/micropython/lib/devlab_ddp.py` module, which is installed as `/lib/devlab_ddp.py` on the host. Low-level scanner sketches are `software/examples/i2c/cpp_examples/i2c_scanner/i2c_scanner.ino` and `software/examples/i2c/micropython/i2c_scan.py`.

5.2 DDP Measurement Transaction

DDP uses command transactions, not a memory-mapped register interface. For each command, the master writes the command byte and issues STOP, waits 2 to 5 ms, then starts a separate read of the exact response length. Multi-byte integers are little endian. A read does not start conversion; it returns the latest background sample.

The mandatory discovery order is `GET_PROTOCOL` (`0x04`), `GET_DEVICE_ID` (`0x00`), `GET_FIRMWARE_VERSION` (`0x01`), `GET_HARDWARE_VERSION` (`0x02`), and `GET_CAPABILITIES` (`0x03`). Only protocol major 1 and Device ID `0x0102` identify this profile. The measurement command `TEMT6000_RAW` (`0x80`) returns two bytes: `raw = response[0] | (response[1] << 8)`, with valid ADC codes from 0 through 4095.

The current responses are Device ID `[0x02, 0x01]` (little-endian `0x0102`), protocol/firmware/hardware `[0x01, 0x00]` (major 1, minor 0), and capabilities `[0xBB, 0x01, 0x00, 0x00]` (little-endian `0x000001BB`). A library should validate these values at startup, cache them, and repeat discovery after reset, address change, or bus recovery.

The byte `0x01` is context-dependent, not a universal status code. It is the `GET_FIRMWARE_VERSION` command when written as a command; version major 1 in `[0x01, 0x00]`; the high byte of Device ID `0x0102` in `[0x02, 0x01]`; a capability-bitmap byte in `[0xBB, 0x01, 0x00, 0x00]`; logical HIGH in a `READ_GPIO0` response; and “address loaded from Flash” in a `GET_I2C_STATUS` response. Interpret it only under the issuing command and expected length.

5.3 Address, Indicator, and Averaged Capture

The common DDP block provides `GET_I2C_ADDR` (`0x20`), `SET_I2C_ADDR` (`0x21`), `SAVE_CONFIG` (`0x22`), `RESET_FACTORY` (`0x23`), and `GET_I2C_STATUS` (`0x24`). Factory address is `0x20`; a new value must be within `0x08..0x77`.

Address change is a staged operation. Send `0x21`, wait/read ACK low nibble `0xD`, send the new one-byte value within 250 ms, wait about 100 ms, and read the final ACK at the old address. The setter saves Flash and reset follows when the final ACK is consumed. Wait about 300 ms and identify `0x0102` at the new address. `SAVE_CONFIG` is currently idempotent. Factory restore returns ACK low `0xE` but requires a separate reset before address `0x20` becomes active.

5.4 Averaged Capture

The controller updates ADC0 in the background about every 20 ms and publishes a rounded circular-buffer mean. `GET_ADC_AVERAGING` (`0x63`) returns 1, 4, 8, 16, or 24. `SET_ADC_AVERAGING` (`0x62`) is staged with ACK low nibble `0xC` and persists to Flash. Read before writing, and wait `N × 20 ms` for a full window. Recommended values are 8 for UI/general automation and 16 for stable ambient measurement. Do not poll faster than 20 ms expecting a new value.

5.5 Built-in Indicator

Controller `PB5/BUILTIN` uses the relay-compatible commands: `RELAY_OFF` (`0xA0`), `RELAY_ON` (`0xA1`), and non-blocking `RELAY_TOGGLE` (`0xA2`). It is not logical GPIO0, and `WRITE_GPIO0/1` are unsupported. Pulse time is configured with `SET_TOGGLE_TIME` (`0xA3`) in `1..40` units of 25 ms.

5.6 Direct Analog Bring-up

1. Leave the I2C/service pins undriven and configure a host ADC as a high-impedance input.
2. Connect the direct `GND`, `VCC`, and `SIG` contacts with power removed.
3. Verify the actual `SIG` range using a multimeter before relying on the ADC; the controller ADC conversion range is `0..VCC`.
4. Compare covered and illuminated readings and check that the response is repeatable.
5. Calibrate against a reference meter for quantitative lux measurements.

The analog examples remain available under `software/examples/adc/`, separate from the I2C clients. Their voltage calculations require V0.3.1 measurement because the only available schematic describes legacy V0.0.1 hardware.

5.7 Modifying the I2C Bridge

Do not cut the bridge as a first troubleshooting step. First verify power, connector order, pull-ups, bus ownership, and scan timing. If analog-only operation requires isolation, remove power, document the original bridge, cut only the marked feature, inspect for debris, and confirm continuity before repowering. Restoring the bridge requires controlled solder rework.

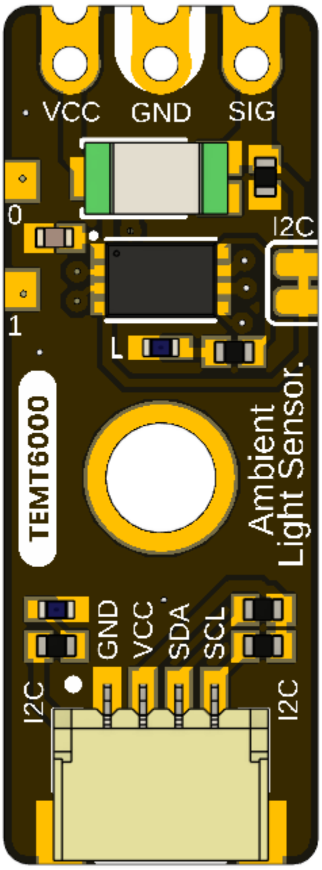
5.8 Troubleshooting Guide

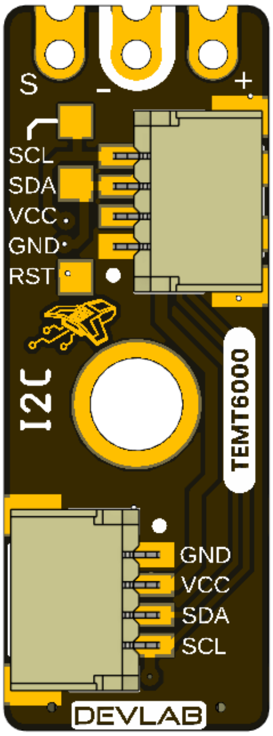
Symptom	Checks
No address found	Supply/bus-logic compatibility, ground, connector orientation, populated connector, SDA/SCL continuity, intact I2C bridge
Bus held low	Cable reversal, duplicate pull-ups, unintended drive on shared `SDA/SWDIO` or `SCL/SWCLK`, solder bridge, unpowered device
Address found but rejected	Verify DDP major 1, Device ID `0x0102`, exact response lengths, and possible address collision
DDP identity succeeds but measurement fails	Check `ANALOG_INPUT/SENSOR_DATA`, 5 ms delay, exact length, byte order, and 12-bit range
Direct `SIG` always zero/full scale	Supply, ADC range, contact order, optical obstruction, current-board transfer function
Built-in LED unexpected	Use relay-compatible commands for PB5; `WRITE_GPIO0/1` are unsupported
Lux result inaccurate	Sensor spread, source spectrum, geometry, enclosure, temperature, ADC and calibration

6 Mechanical Information

The current package provides rendered top and bottom views for V0.3.1 but no controlled current-revision dimension drawing. Legacy V0.0.1 dimensions are archived under `hardware/resources/old/` and describe a materially different, shorter analog-only board.

6.1 Current Board Envelope





The board has a narrow module outline, direct contacts at one end, at least one central mounting feature, and optional horizontal Qwiic connector positions. Rendered views are suitable for identification, not production measurement.

6.2 Integration Clearances

Leave optical clearance around the transparent TEMT6000 sensor package and its field of view. Provide connector insertion and cable bend clearance at every populated Qwiic position. Keep conductive fasteners away from pads and service signals, and avoid enclosure features that shadow the sensor or LEDs.

6.3 Mechanical Data Not Specified

- Board length, width, thickness, tolerances, and finished mass
- Hole diameters, coordinates, and edge clearances
- Optional connector population variants and mating-part numbers
- Connector insertion envelope, retention force, and cable bend radius
- Maximum top- and bottom-side component heights
- Optical axis, keep-out cone, and enclosure-window specification
- PA0/PA1 and service-pad coordinates

Do not scale production dimensions from the images or reuse the archived V0.0.1 mechanical values for V0.3.1.

7 Company Information

Item	Information
Company / brand	UNIT Electronics
Board ecosystem	DevLab
Product family	Atom
Website	UNIT Electronics
Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX, Mexico

The product repository and controlled technical references are listed in the next chapter.

8 Reference Documentation

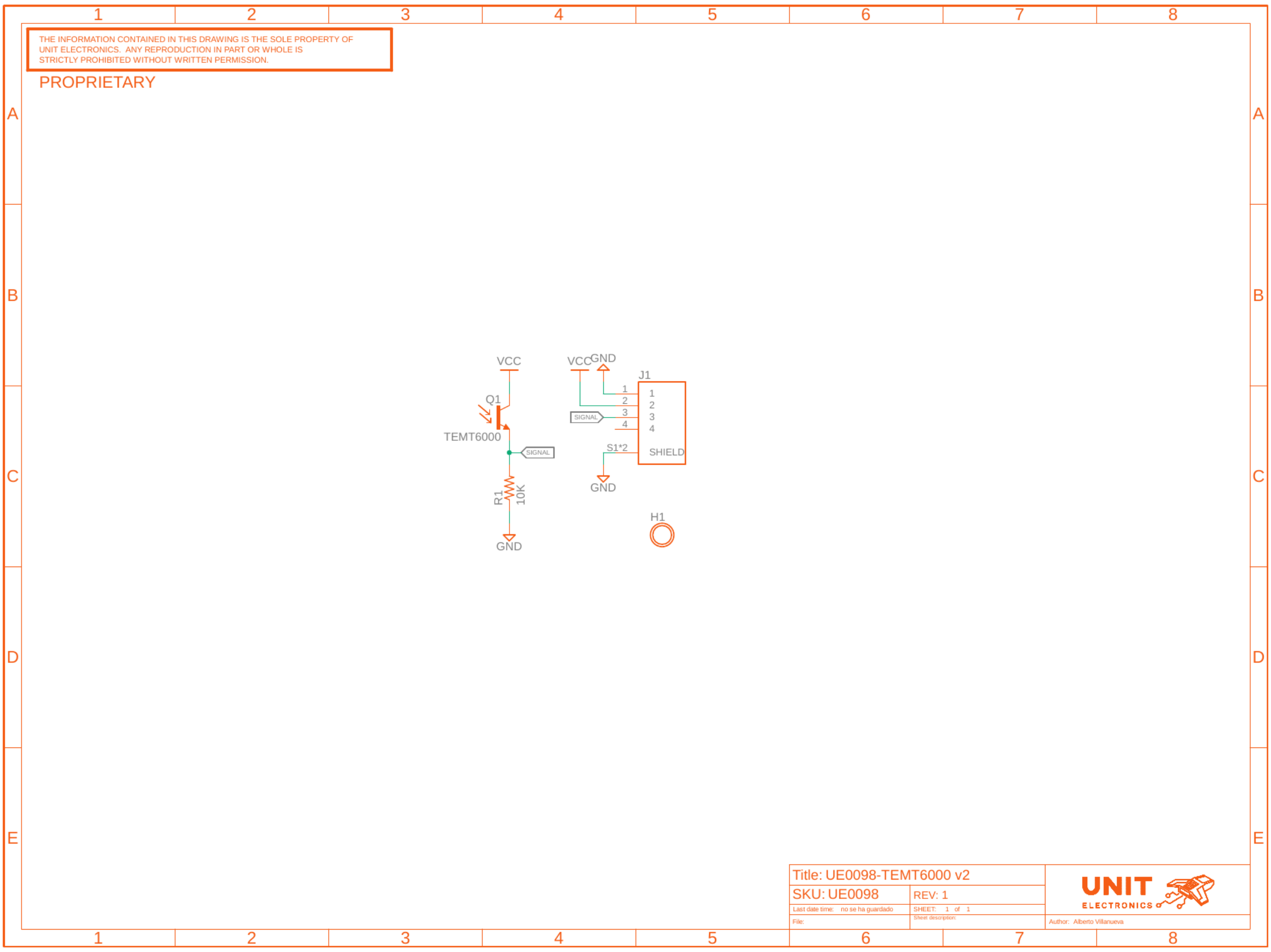
Reference	Location
Product repository	UNIT-Electronics-MX/unit_devlab_temt600_ambient_light_sensor
Hardware reference	hardware/README.md
English pinout	Released pinout — English
Spanish pinout	Released pinout — Spanish
Board top view	V0.3.1 resource
Board bottom view	V0.3.1 resource
Legacy schematic	Analog V0.0.1
Software examples	C++ and MicroPython
Local DDP device profile	TEMT6000 DDP profile
Canonical DDP definitions and Arduino library	UNIT-Electronics-Labs/unit_devlab_ddp_library
Comparative component page	Vishay TEMT6000X01 — reference only
Comparative component datasheet	Vishay document 81579 — reference only

The released pinout and current board images define the visible interfaces. The legacy schematic defines only the earlier analog module. The Vishay datasheet is retained only as a comparative TEMT6000X01 profile; it does not establish the manufacturer or orderable part fitted to this module and does not define complete-module ratings. The DDP profile defines runtime identity and digital data transport; it does not replace the missing V0.3.1 electrical specification.

9 Appendix

9.1 Legacy V0.0.1 Schematic

The image below is rendered from `hardware/unit_sch_V_0_0_1_ue0098_TEMT6000.pdf`. It documents the former analog-only module and is retained for traceability. It is **not** the electrical schematic for the controller-based V0.3.1 board.



The legacy circuit connects a TEMT6000 phototransistor and 10 kΩ load to one analog signal. It contains no interface controller, SDA, SCL, PA0, PA1, reset, SWD, indicators, or I2C disable bridge.

9.2 Document Control

Field	Value
Product	UNIT ATOM TEMT6000 Ambient Light Sensor
Product hierarchy	UNIT Electronics → DevLab board ecosystem → Atom family
Manufacturer Part Number (MPN)	UE0098
Current board artwork	V0.3.1

Field	Value
Current pinout	V3.1.0
Available schematic	Legacy V0.0.1 only
Product Reference	Version 1.1.0
Publication date	2026-08-31
Interfaces	Qwiic I2C with shared factory SWD, analog, and auxiliary pads

9.3 Required Technical Releases

- V0.3.1 schematic and bill of materials
- Exact controller memory/package option, released firmware image, and update procedure
- Exact oscillator/timing tolerances and uniform future DDP status behavior
- Complete-module supply/logic absolute-maximum ratings and current consumption
- Pull-up values and bus-loading limits
- Direct `SIG` transfer, loading, range, and ADC guidance
- Current-revision controlled mechanical drawing
- Module-level optical, electrical, environmental, and EMC characterization

9.4 Source Notes and Inconsistencies

- Current board images and pinout resources use different revision-number formats in their filenames and storage directories.
- Current pinout artwork calls the product `DevLab: I2C TEMT6000`; this reference identifies UNIT Electronics as the company that creates and develops the DevLab board ecosystem, with Atom as a product family within it.
- The current pinout shows an onboard controller and I2C/debug resources, but the only schematic in the repository is the earlier analog V0.0.1 circuit.
- Archived V0.0.1 dimensions, topology, and board views do not describe the longer V0.3.1 controller-based board.
- Qwiic positions A and B are marked optional; released assembly variants and controlled contact numbering are not supplied.
- Earlier examples treated the sensor as a digital threshold input. The sensor signal is analog; I2C operation is provided by the new onboard controller.
- Current firmware maps PA2 to `ADC0`, PA4 to read-only `GPIO0`, and PB5 to the relay-compatible actuator block. Digital output commands `0x42`/`0x43` are not implemented; exposed `PA0`/`PA1` remain unassigned.

9.5 Current Firmware Limitations

- The `WATCHDOG` capability is announced, but the current watchdog command does not operate an active hardware IWDG and must remain experimental.
- `GET_RESET_INFO` is not implemented and NRST behavior is incomplete.
- An internal ADC error can be returned as `0x0FFF`, indistinguishable from a legitimate full-scale sample.
- Current main-loop code does not call the available prolonged-BUSY recovery.
- There is no sample timestamp, sample counter, or calibrated lux result.